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EC39002

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Experiment 1: Line Coding



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1 Aim of the Experiment

The objective of this experiment is to design a circuit for the generation of different line coding schemes for a given PN sequence. Specifically, we aim to implement and observe:

1. Unipolar Return-to-Zero (RZ)
2. Manchester Coding
3. Bipolar Return-to-Zero (Bipolar RZ)
4. Alternate Mark Inversion (AMI)
5. Multi-Level Transmit (MLT-3) (Bonus)

2 Components Used

The following components were used to design the line coding circuits:

- **ICs:** IC 7486 (XOR), IC 7404 (NOT), IC 7408 (AND), IC 7474 (D FlipFlop), IC 741 (Op-Amp) ,IC 7432(OR gate)
- **Equipment:** Digital Storage Oscilloscope (DSO), Function Generator, $\pm 12V$ DC Power Supply.
- **Miscellaneous:** Breadboard, resistors, capacitors, and connecting wires.

3 Theory

Line coding is used for the transmission of digital data over a communication channel by representing binary digits (0, 1) as electrical waveforms. The choice of code involves trade-offs regarding bandwidth, DC component, and clock recovery.

3.1 Unipolar RZ

In Unipolar RZ signaling, a logic '1' is represented by a pulse that is active for half the bit duration ($T_b/2$) and returns to zero for the remaining half. A logic '0' is represented by the absence of a pulse.

3.2 Manchester Coding

Manchester coding is a self-clocking scheme where each bit period has a transition. A logic '1' typically involves a transition from High to Low (or Low to High depending on convention) in the middle of the bit interval. This ensures zero DC component.Clock Recovery is easier .

3.3 Bipolar RZ

In Bipolar signaling, three voltage levels ($+V, 0, -V$) are used. The non-zero pulses return to zero voltage for the second half of the bit duration.

3.4 AMI

Logic '1's are represented by alternating positive and negative pulses. Logic '0' is represented by zero potential.

3.5 MLT-3

In this coding voltage level cycles from (0 to +5 to 0 to -5 to 0 ...) if bit 1 is encountered and it remains same if bit 0 is encountered .

4 Circuit Implementation

4.1 Data Synchronization

Before line coding, the data is synchronized with the clock using a D-Flip Flop to ensure stable timing.

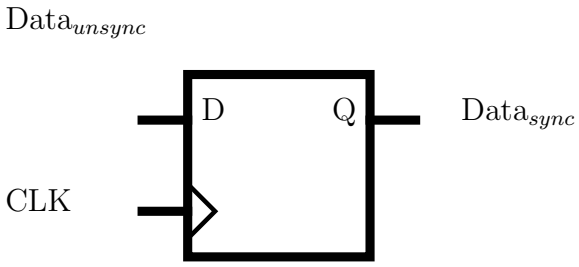


Figure 1: Data Synchronization Circuit

4.2 Unipolar RZ Circuit

The Unipolar RZ signal is generated by ANDing the synchronized data with the Clock signal.

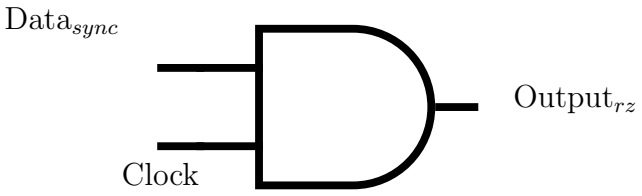


Figure 2: Unipolar RZ Schematic

4.3 Manchester Circuit

Manchester encoding is achieved using an XOR operation between the Data and the Clock (often followed by an inversion depending on the desired convention).



Figure 3: Manchester Encoding Schematic

4.4 Bipolar RZ Circuit

The Bipolar RZ circuit utilizes logic gates to separate the data stream and an Op-Amp to create positive and negative swings.

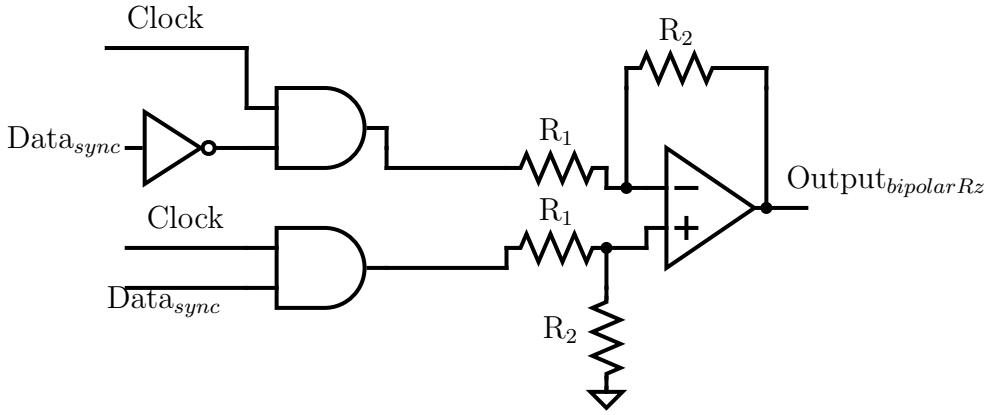


Figure 4: Bipolar RZ Schematic

- Output Voltage of Differential Ampilfier is $V_{out} = \frac{R_2}{R_1} * (V_+ - V_-)$

4.5 AMI Circuit

The AMI circuit uses a flip-flop and logic gates to steer alternate '1's to different paths, summed by an Op-Amp to produce alternating polarity.

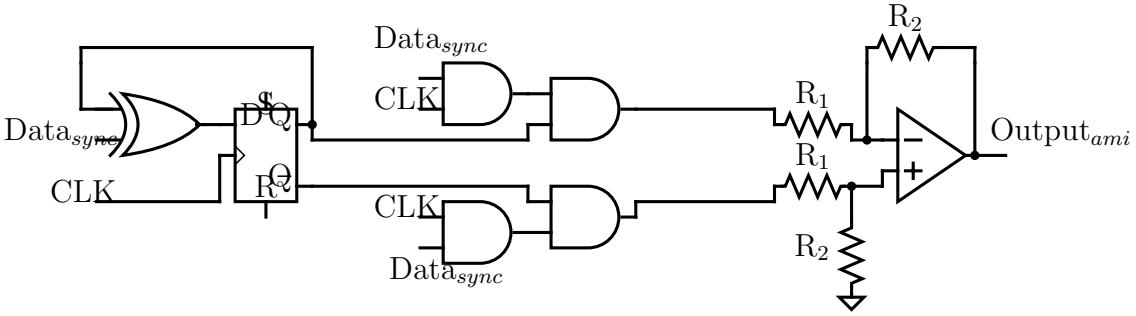


Figure 5: AMI Schematic

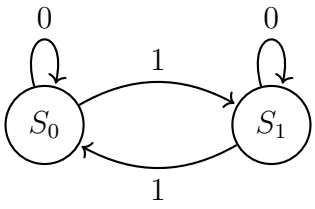


Figure 6: State Transition Diagram of AMI Encoder.

- S_0 = State represents last high was +5 V
- S_1 = State represents last high was -5 V

4.6 MLT-3 Circuit

MLT-3 cycles through three voltage levels (-1, 0, +1, 0) upon every logic '1'.

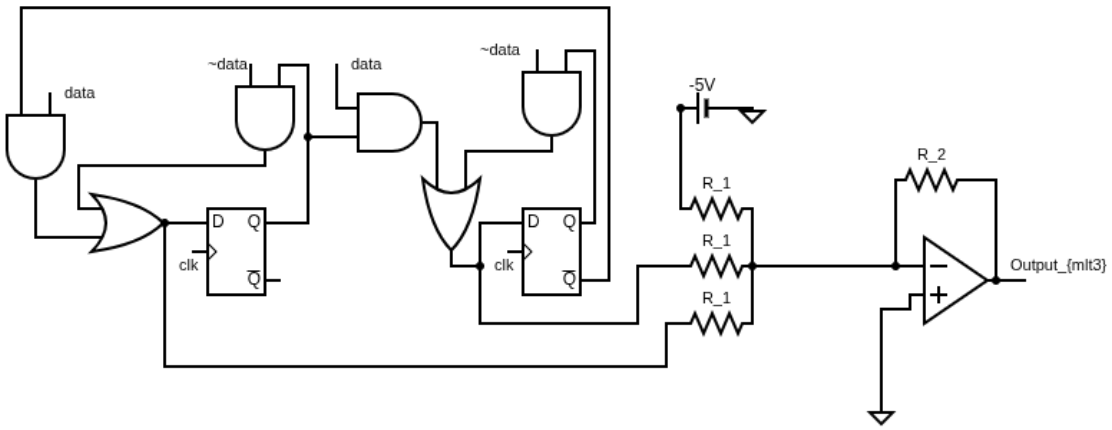


Figure 7: MLT-3 Schematic

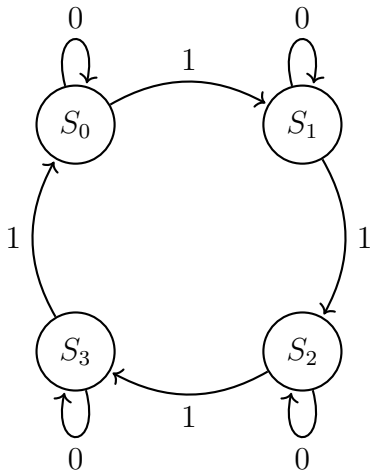


Figure 8: State Transition Diagram of MLT-3

- S_0 = State represents state of -5V
- S_1 = State represents state of 0V coming from -5V
- S_2 = State represents state of +5V
- S_3 = State represents state of 0V coming from +5V

5 Observations and Results

5.1 Unipolar RZ Waveform

The output waveform shows the pulse returning to zero amplitude halfway through the bit period for logic '1'.

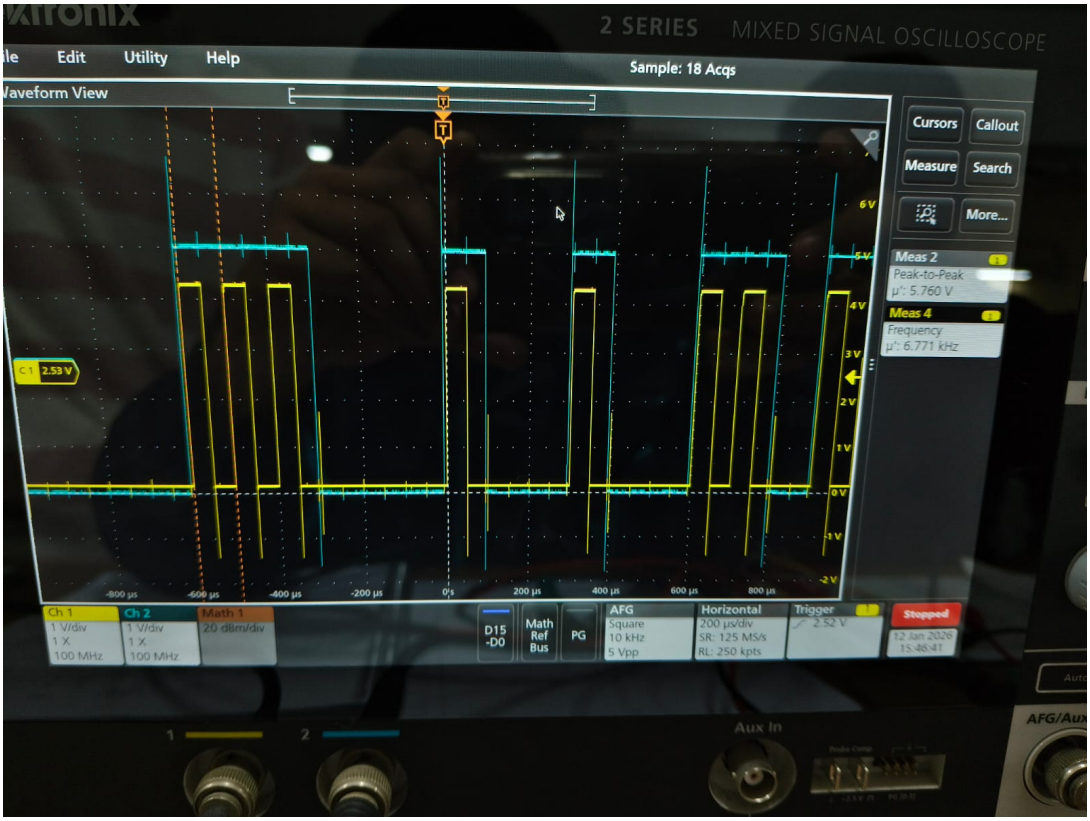


Figure 9: Unipolar RZ Output (Yellow) vs Input Data (Cyan)

5.2 Manchester Waveform

The waveform demonstrates the characteristic mid-bit transition for every data bit, ensuring synchronization capabilities.

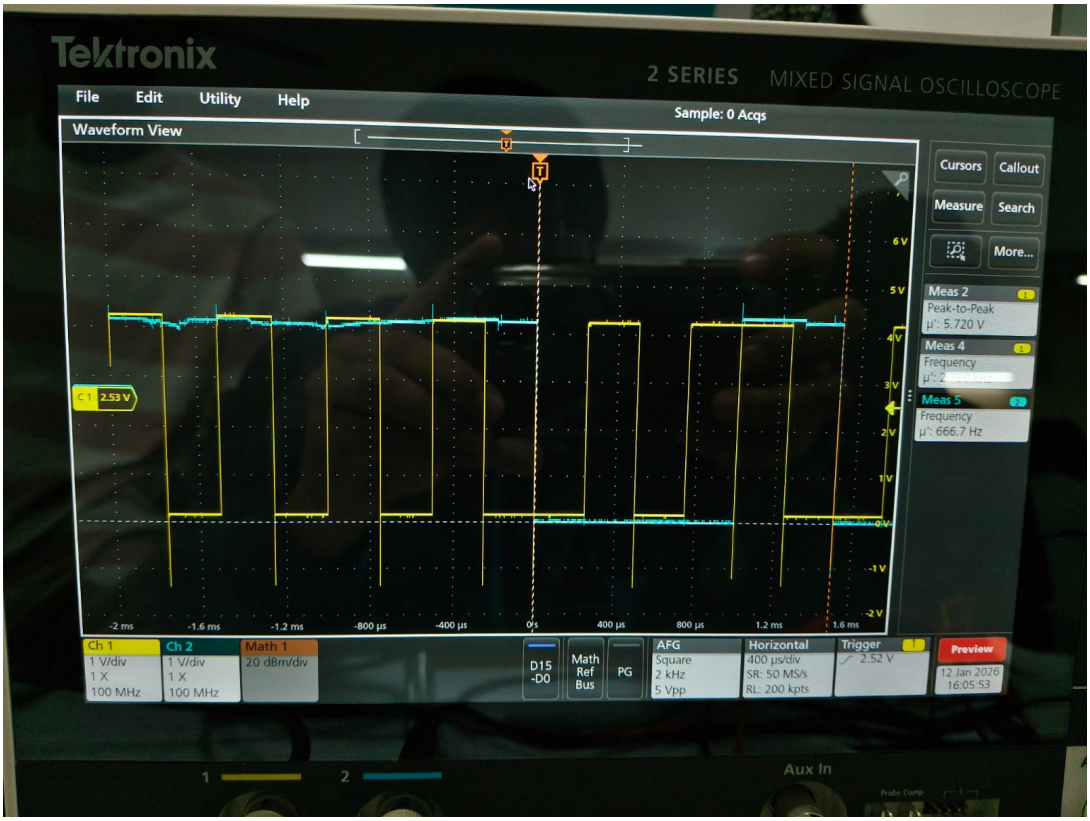


Figure 10: Manchester Coding Output(Cyan) vs Data(Yellow)

5.3 Bipolar RZ Waveform

The output shows pulses , +5V if bit is 1 and -5V if bit is 0 , And also returning back to zero mid-bit .

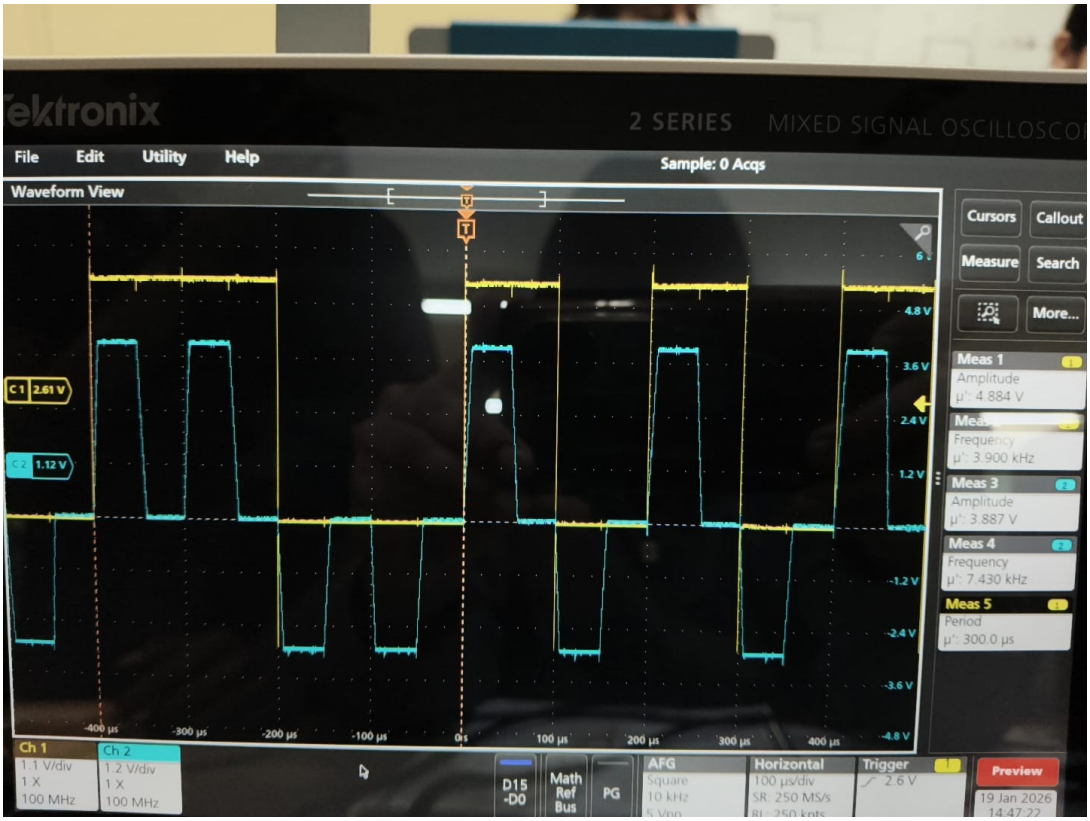


Figure 11: Bipolar RZ Output

5.4 AMI Waveforms

The Alternate Mark Inversion is observed where logic '1' alternates between $+V$ and $-V$, and logic '0' stays at ground.



(a) AMI Output 1(NRZ



(b) AMI Output 2(RZ)

Figure 12: AMI Waveforms showing alternating polarity for Logic 1

5.5 MLT-3 Waveform

The waveform transitions to the next level in the sequence (0 → +V → 0 → -V) only when a logic '1' occurs.

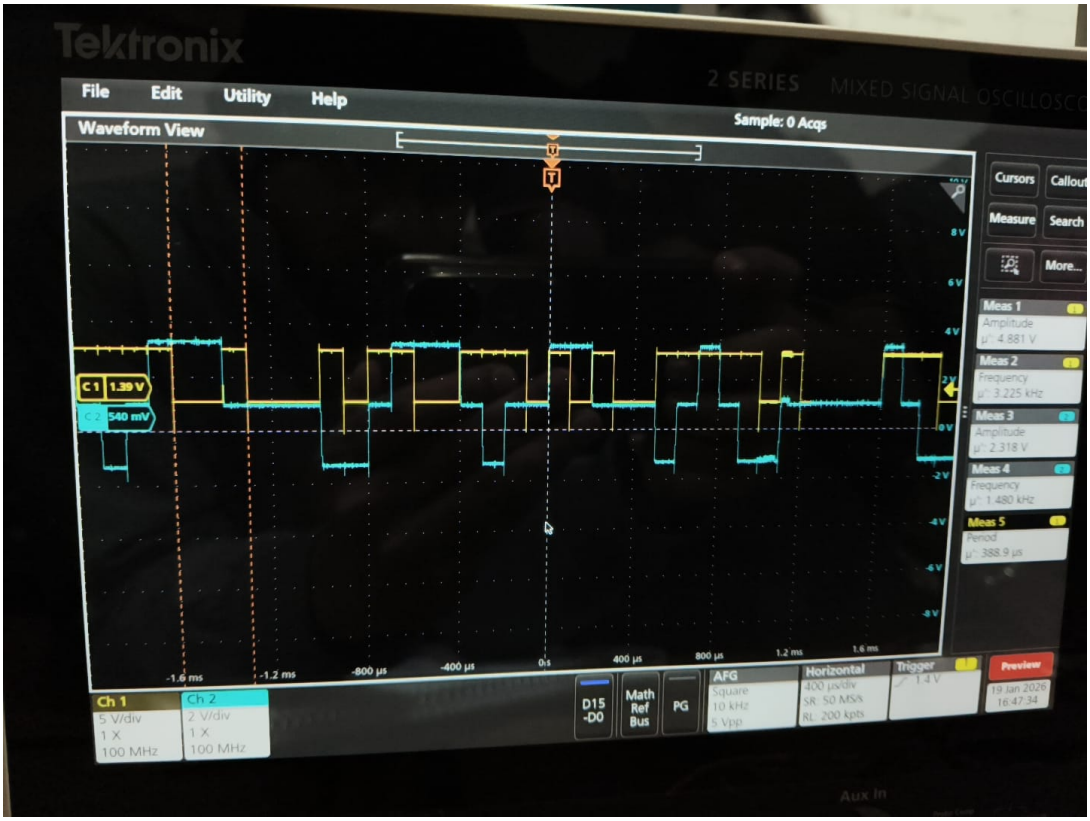


Figure 13: MLT-3 Output Waveform

6 Discussion

- **Data Synchronization:** The input data stream was sourced from an external signal generator, while the clock signal was generated by the oscilloscope's internal AFG. This setup introduced a probability of phase desynchronization between the data and clock. To resolve this, a D Flip-Flop was introduced at the input stage to synchronize the data stream with the system clock before line coding.

- **Unipolar RZ Implementation:** The Unipolar Return-to-Zero (RZ) coding was the most straightforward to implement. It was achieved by performing a logical `AND` operation between the synchronized Data stream and the Clock signal, ensuring the output pulse duration matches the clock's high duty cycle.

- **Manchester Coding Advantage:** Analysis of the Manchester waveform confirms its primary advantage: self-clocking. Since a transition is guaranteed to occur in the middle of every bit period, the clock signal can be recovered directly from the incoming data stream, eliminating the need for a separate clock synchronization line.

- **Bipolar RZ Voltage Levels:** A major challenge in Bipolar RZ was generating three distinct voltage levels $(+V, 0, -V)$, which is not possible using standard binary logic gates alone. This was solved by using a combinational logic circuit to separate positive and negative pulses, followed by a Differential Amplifier. The amplifier synthesizes the final output by performing the operation $V_{out} = V_2 - V_1$.

- **AMI Logic and Memory:** The Alternate Mark Inversion (AMI) scheme required memory of the previous logic '1' state to alternate polarities. This was implemented using a D Flip-Flop as a storage element combined with combinational logic and a final Differential Amplifier.
  - For **AMI-NRZ (AMI-1)**, the clock input to the logic gates was tied high, creating full-width pulses.
  - For **AMI-RZ (AMI-2)**, the clock input was connected to the system Clock (similar to Unipolar RZ), ensuring the signal returns to zero within the bit duration.
- **MLT-3 State Machine & Offset:** The MLT-3 encoding requires cycling through 4 internal states to produce 3 voltage levels. This was realized using two D Flip-Flops (representing states D_1, D_2) and a Summing Amplifier. The amplifier summed the logic outputs with a $-5V$ reference to shift the levels. **Observation:** A non-zero DC offset was observed during the "zero" voltage state (ground). This is attributed to slight tolerance mismatches in the resistor values used in the summing amplifier circuit, preventing perfect cancellation to exactly $0V$.